## Potentially needed Features:

- Switching Camera from first to third-person camera
- Move with the character
- rotate the camera

## Open Questions / possible Solutions:

**When to switch camera**
- Anytime via Scrollwheel
- Anytime via Button & Scrollwheel -> how to move camera back to player perspective from godmode cam & where does godmode cam get placed upon switching to it
    - Button that Lerps between camera positions
    - Timer in the background, that while in player perspective, lerps the godmode cam back to be behind player
    - Timer, that after a certain time (in player perspective) resets the godmode cam
    - map positions have a declared starting godmode cam position
- Map-Locations
    - how to split them
    - map positions have a declared starting godmode cam position
- via Item
    - where to collect

**How to Rotate the camera**
- Fixed behind the character
- Added up & down rotation -> how far up & down?
    - limit by 0° (=ground) and 90° (directly from the top) (or some other limitation)
    - completely around
- Left-Right rotation independent of character -> How to move character
    - W/S to move along surface (in the direction the player is facing) & additional keys for rotating the character around
    - Added keys for rotating the character to different directions
    - Character rotates head towards the direction the camera is facing, but not their body
- Fixed on the current shape -> how to confuse & keep rotation
    - maybe the fact that player can walk onto shapes changes how they perceive the rotation

## Order of trying solutions:

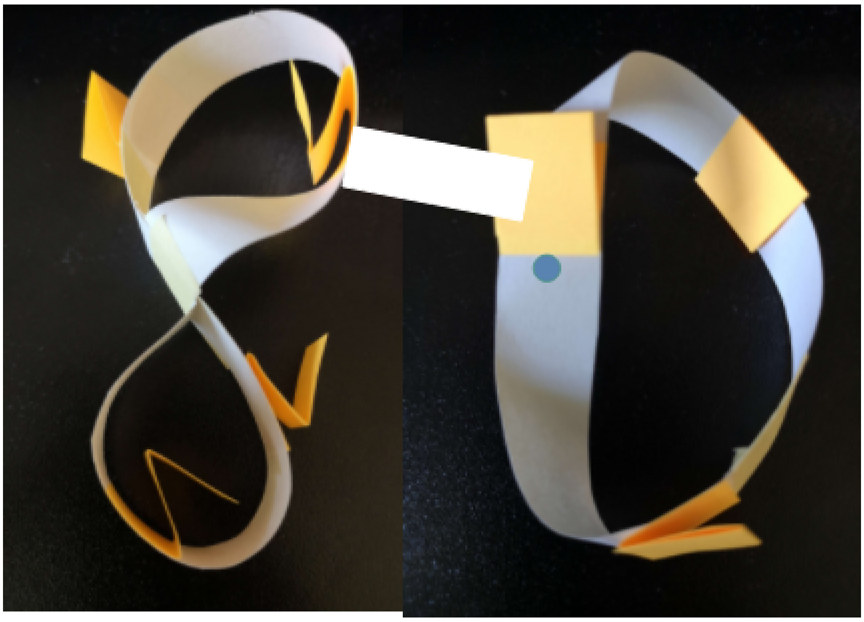
**First solution**
    - Scrollwheel
    - reset to godmoden-cam button
    - resimplify movement (= character can only be rotated along the strips; at crossing the character there is a rotate-around button)
    - fix on current shape -> walking over bridges lerps to next position

**Improvements that should be tried**
    - keeping two switchable cams

- Map-Points

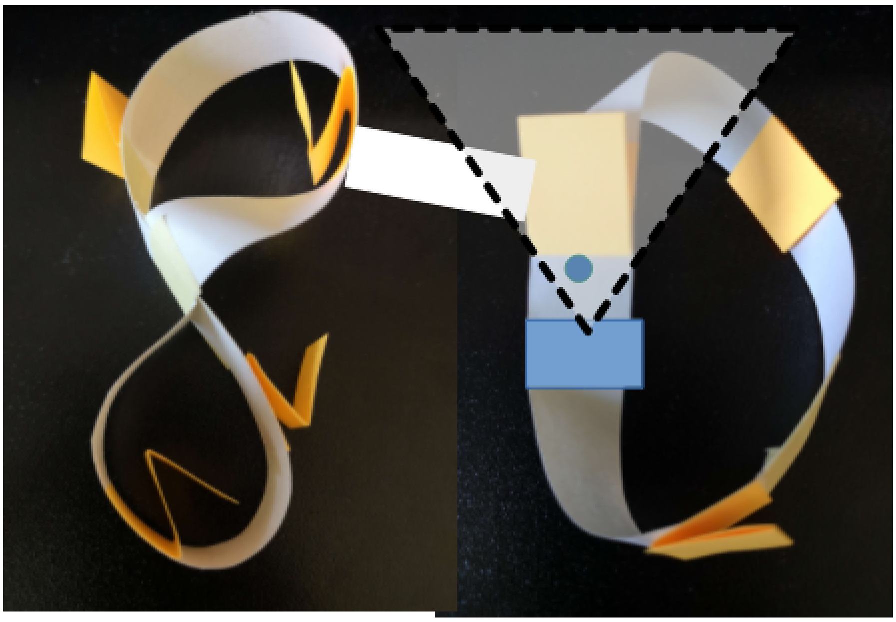
## 2 Camera Modes:

**Godmode Camera** shows all entangled shapes at once:



**Character Camera** shows only a part of the whole thing;
There are two options for this; I'd like to try both (=pls create option A first, afterwards option B with a public bool that let's us switch between the options during playtest):

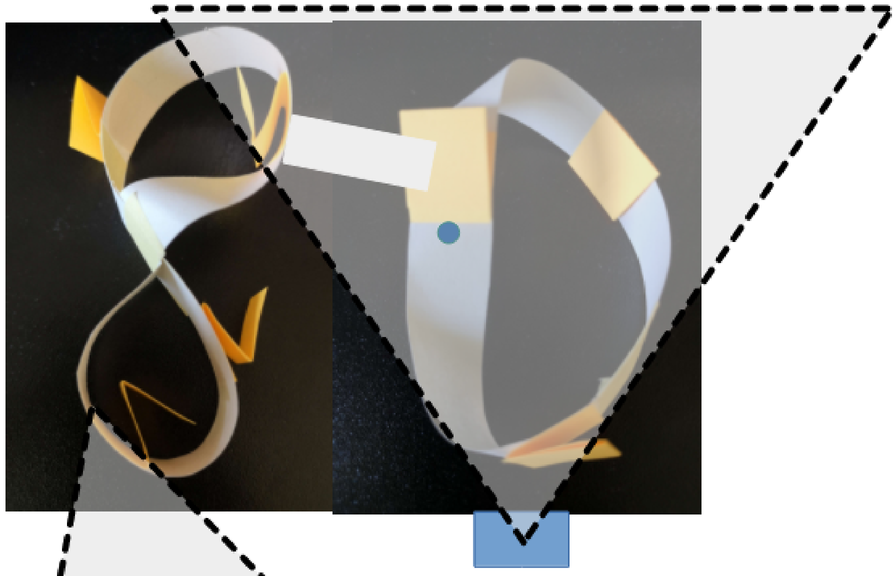
**Option A, Character zoomed:**



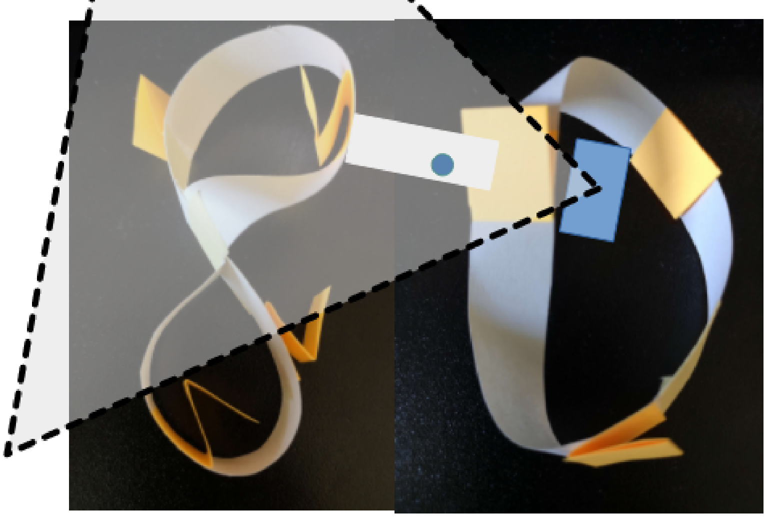
Camera closely behind the character (pls softcode the distance)

Character turns / moves = camera follows

**Option B, Shape zoomed:**

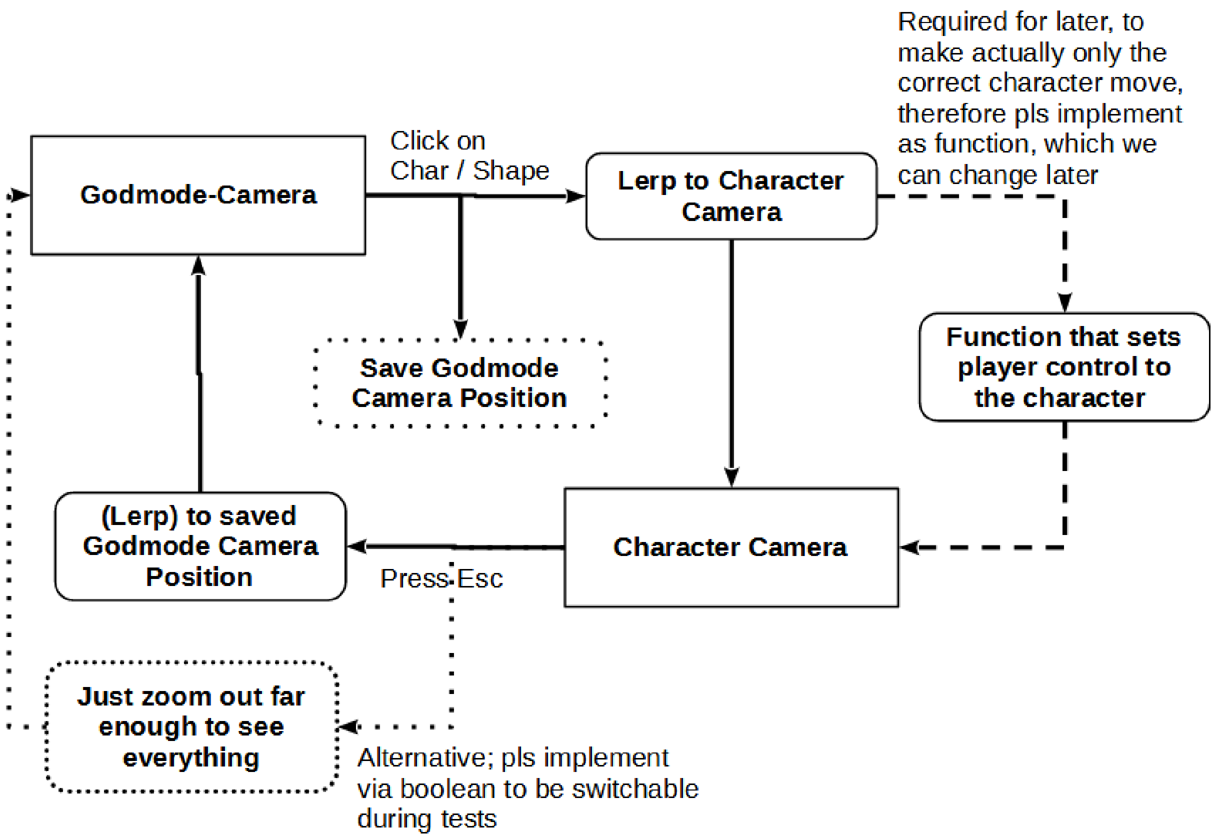


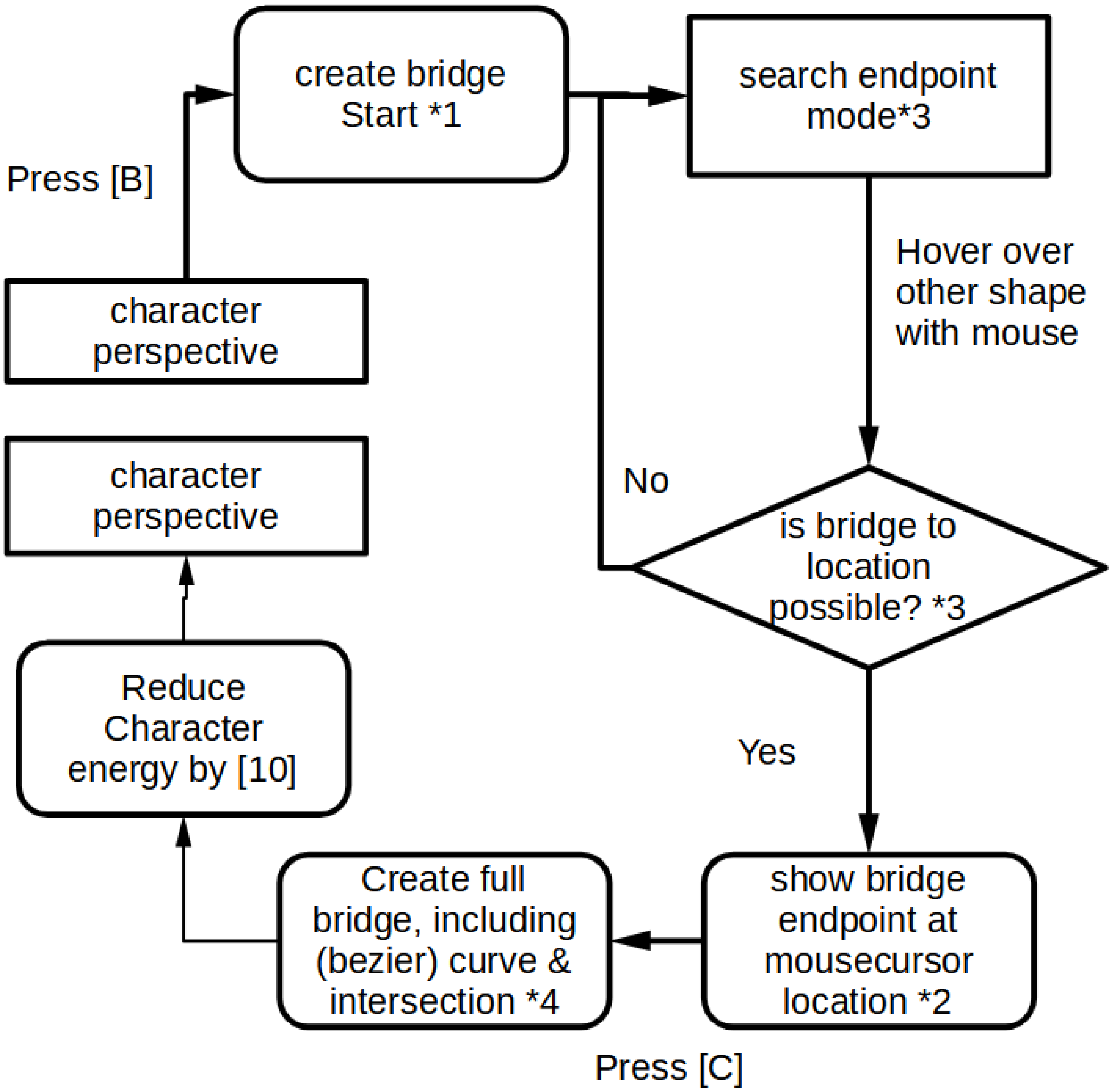
The camera zooms in, so that it sees the whole shape at once

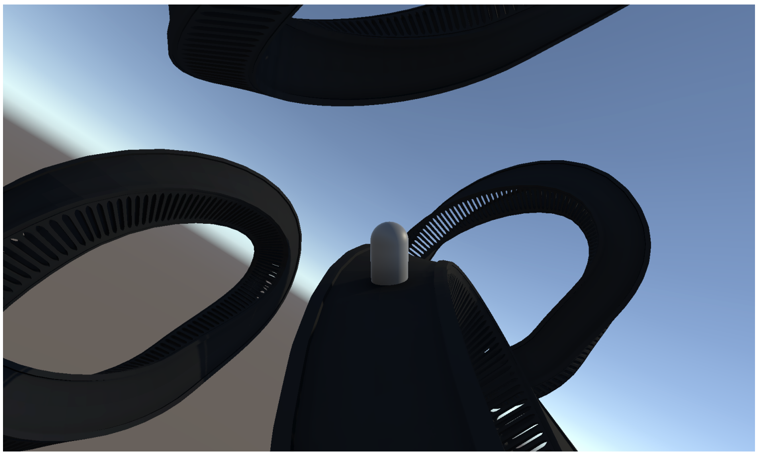


Character walks onto bridge / other shape -> Camera moves/rotates to new position to see the bridge/other shape

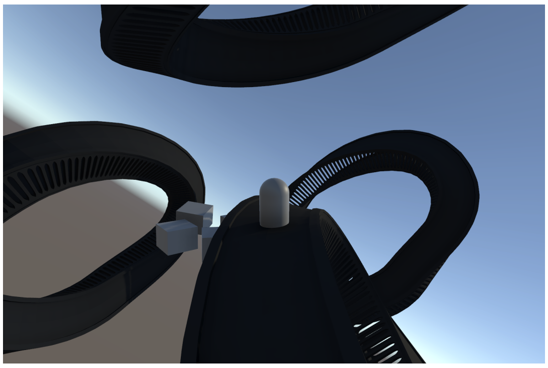
**Switching Cameras**



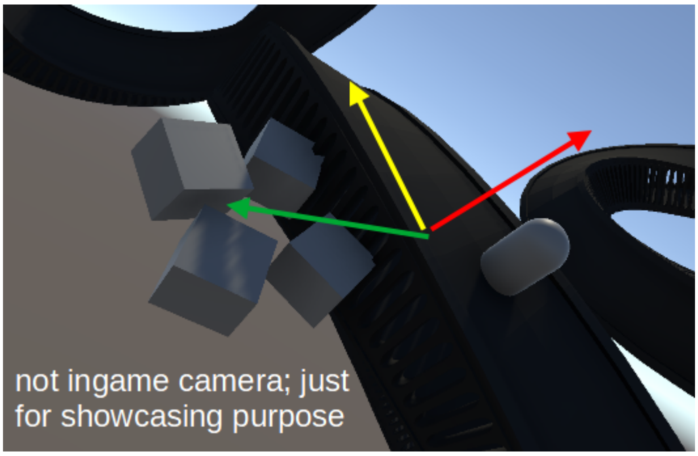

create bridge Start *1
search endpoint mode*3
Press [B]
Hover over other shape with mouse
character perspective
character perspective
No
is bridge to location possible? *3
Reduce Character energy by [10]
Yes
Create full bridge, including (bezier) curve & intersection *4
show bridge endpoint at mousecursor location *2
Press [C]



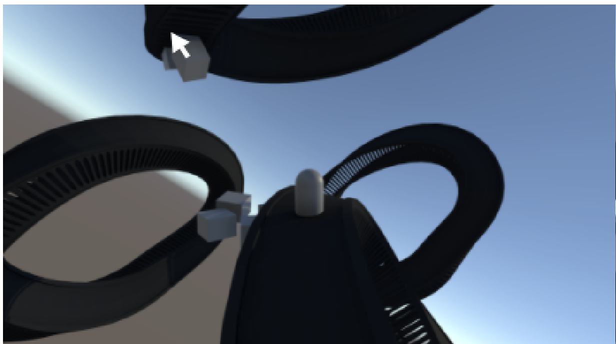
*0: The player takes control over a character
-> Character Camera (see camera document)

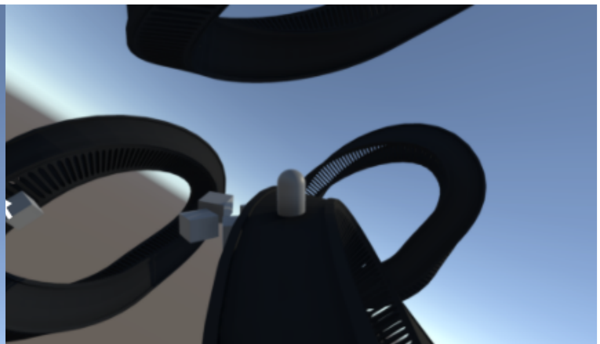


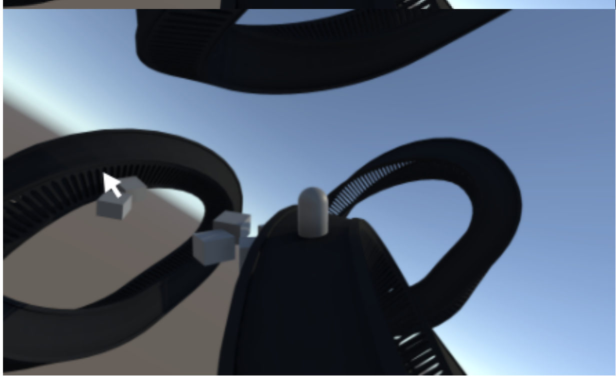
*1 The Player starts building a bridge;
The start-point & end-point of bridges are going sideways off the path



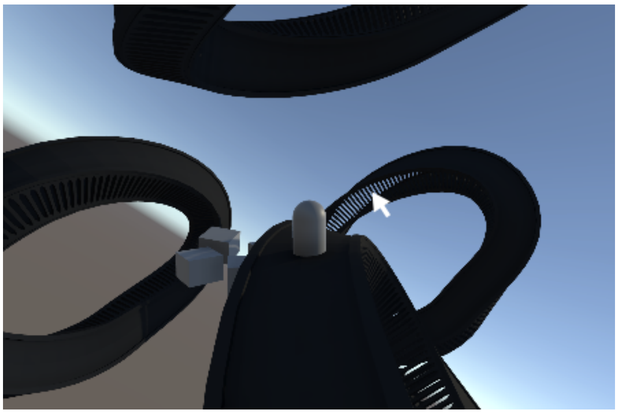

Red = normal of the curve
Yellow = direction the curve leads to
Green = orthagonal to the other two => starting vector of the bridge



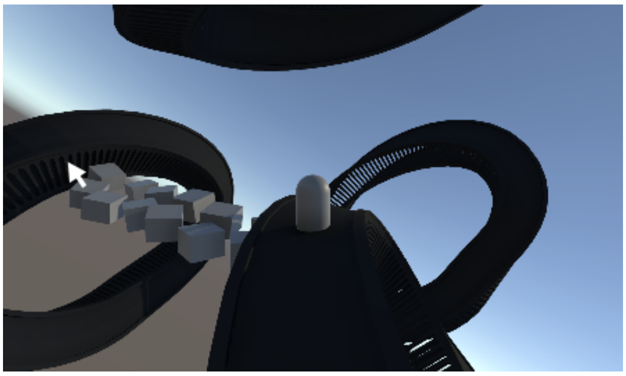




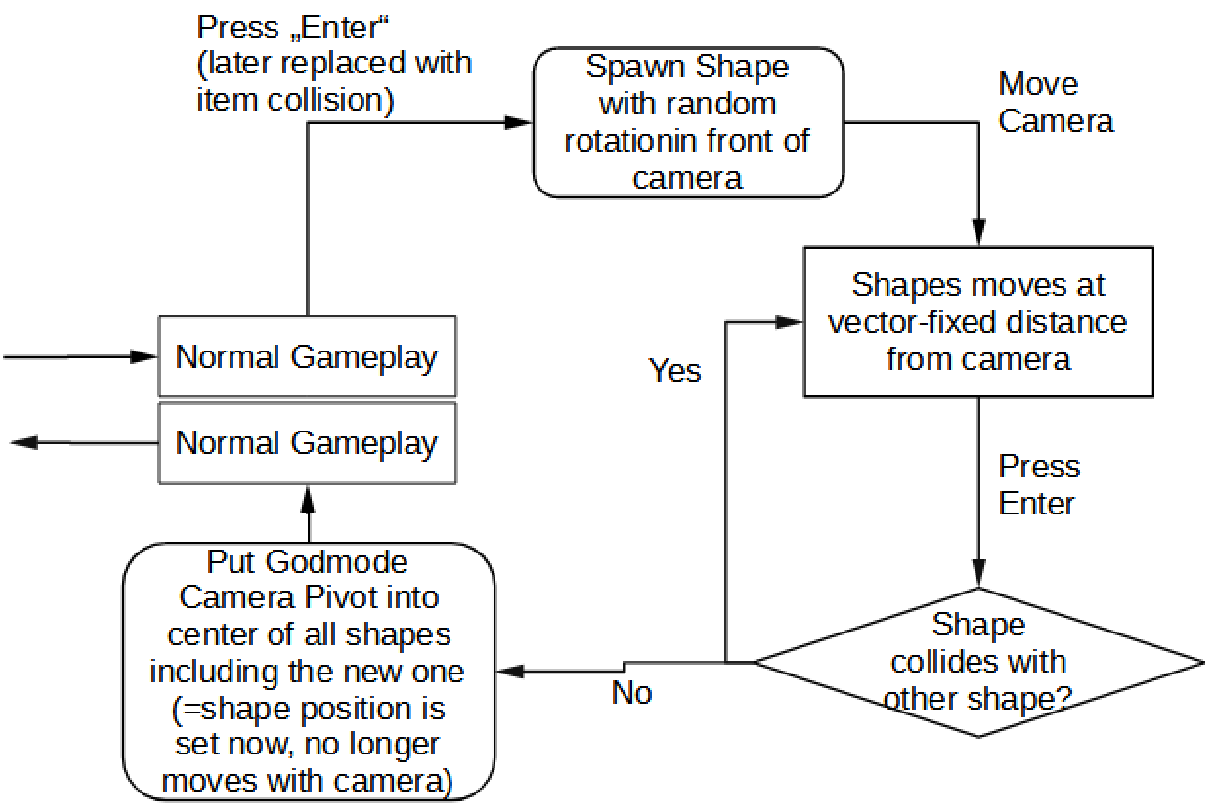
*2 Hovering with the mouse shows bridge end locations

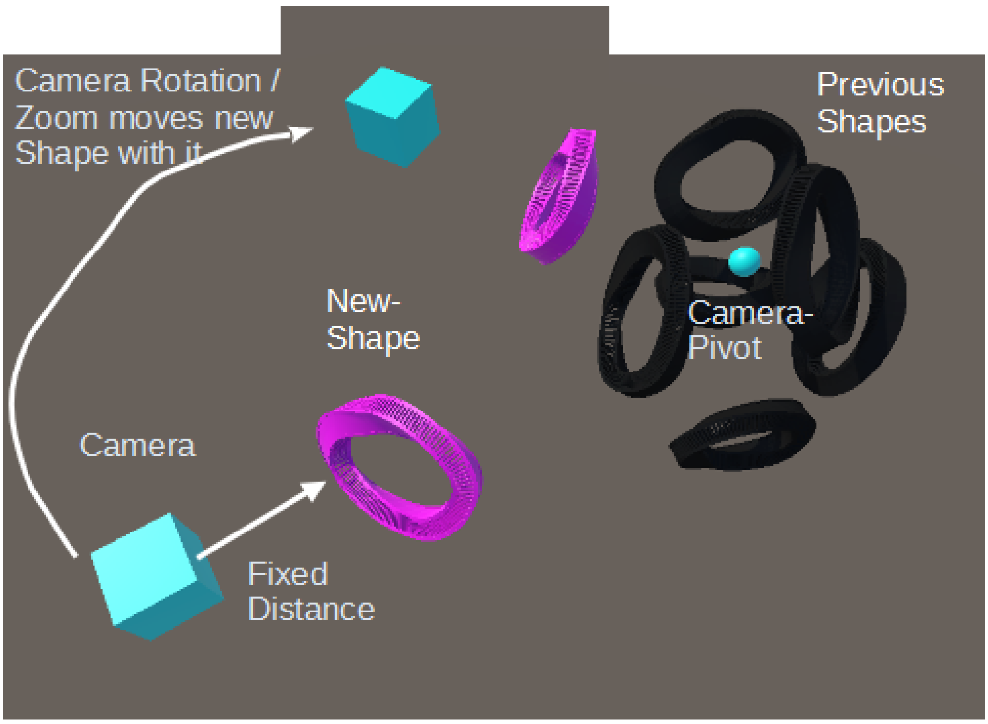


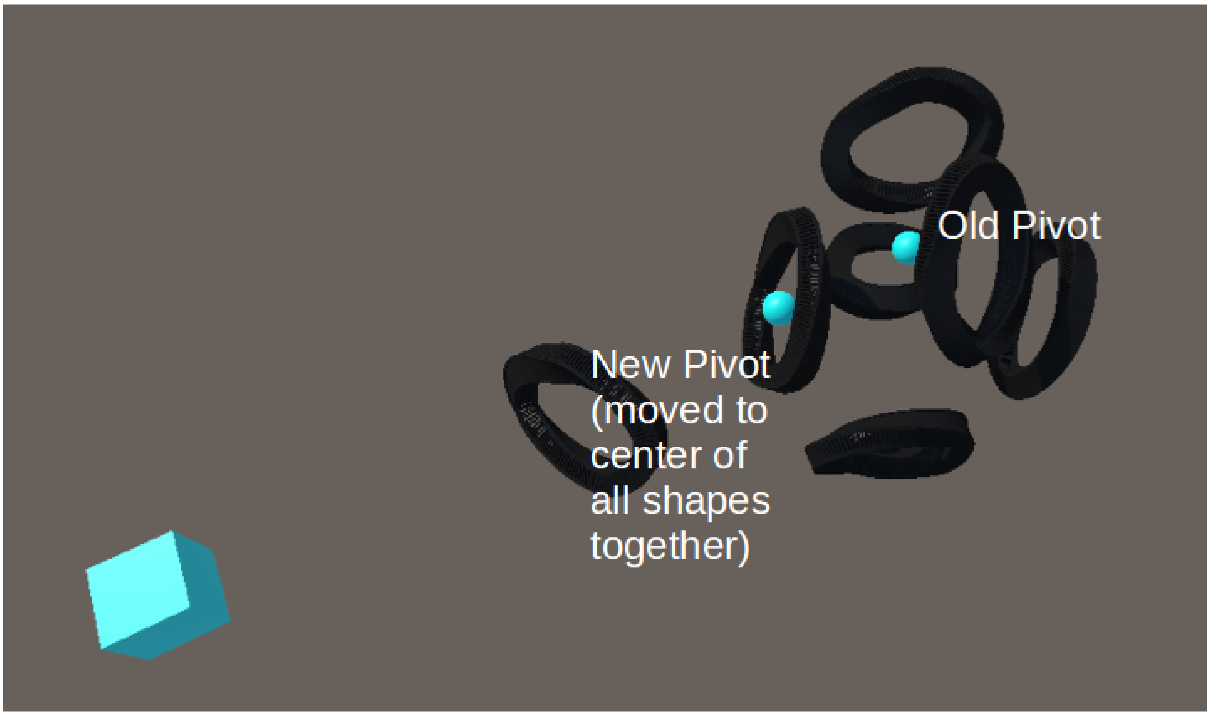
*3 Some location (e.g. too far away; or tilted in a too different direction [=whatever is too hard to implement]) can't be reached with a bridge, so no other end of the bridge is foreshadowed



*4 Creating a bridge = creating a curve & some objects along (& directly next to) the curve, that are facing according to the normals of the bridge at that point


Press „Enter“
(later replaced with
item collision)
Spawn Shape
with random
rotationin front of
camera
Move
Camera
Shapes moves at
vector-fixed distance
from camera
Yes
Normal Gameplay
Normal Gameplay
Press
Enter
Put Godmode
Camera Pivot into
center of all shapes
including the new one
(=shape position is
set now, no longer
moves with camera)
No
Shape
collides with
other shape?


Camera Rotation /
Zoom moves new
Shape with it
Previous
Shapes
New-
Shape
Camera-
Pivot
Camera
Fixed
Distance


Old Pivot
New Pivot (moved to center of all shapes together)

“new-shape item” what happens upon collision:

- loss of energy (due to creating a new shape)
- gives bridges (requires bridges as items)

Ways to revive dead shapes:

- collecting an energy-item shortly before it expires allows to revive a shape
- Two characters colliding on a dead shape revives the dead shape

energy-related abilities:

- Ability to switch from one to another character for x-energy
- Ability to switch & transfer energy from char X to char Y (if in -sight)
- merging a character with an energy-item creates a building, that continuously gives energy (requires the player themselves to have energy

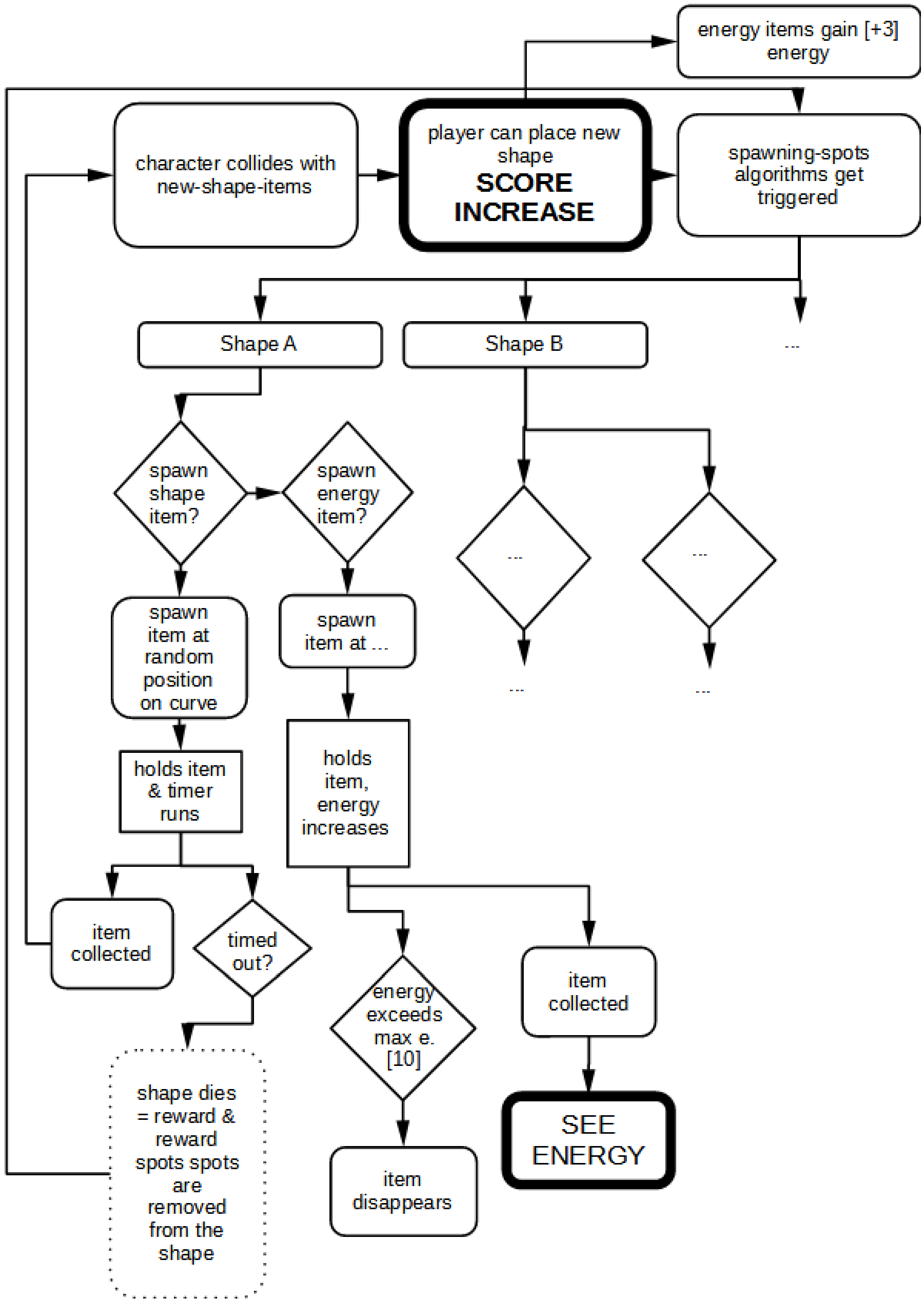
Energy-item-related:

- When full energy, blow a whole into the ground, which pulls characters to the other side (or lets character switch to the other side here (for x-energy))
- upon expiring turns into a dead-energy cell, that a character can merge with to create a permanent energy source
- upon expiring can be transformed into a temporary character
- character getting teleporter to exploding energy

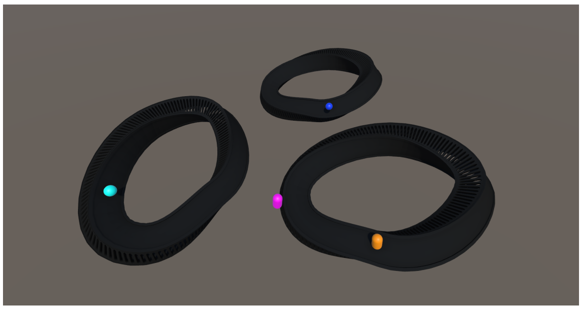
Landmark related:

- dead characters turning into landmarks

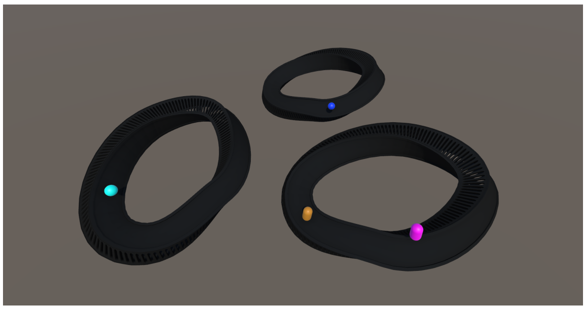
There are two types of items: “shape-creation-items” and “energy items”;
They can be spawned on so-called reward-spots according to the following resource flow:



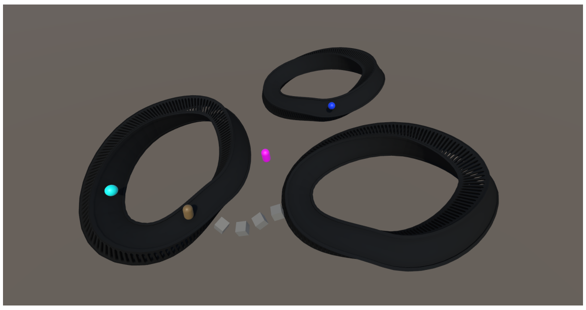
Gameplay Example:



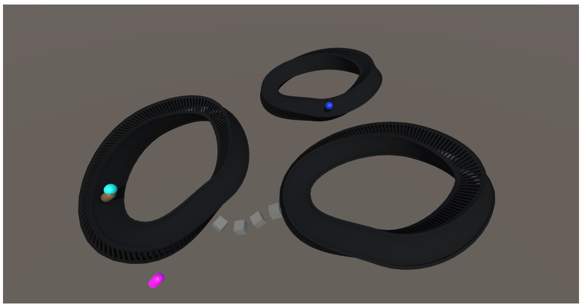
The game start:
Each shape checks whether it should spawn an energy- or new-shape-item



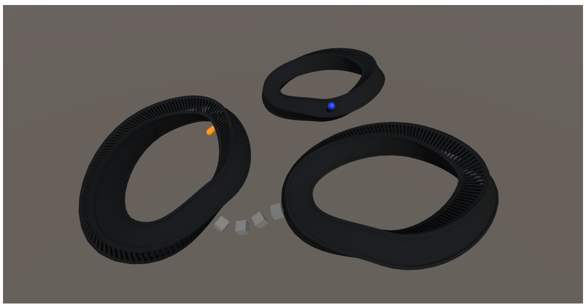
the player takes control over the character (loss of energy) turns it around (small loss of energy)



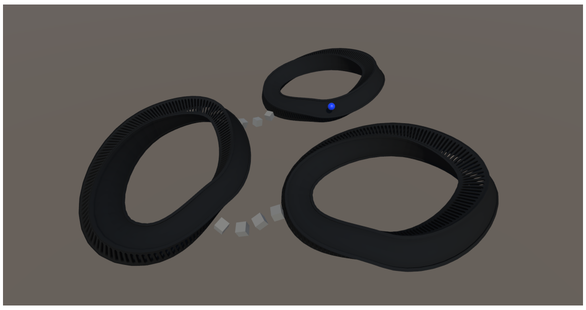
The player builds a bridge (big loss of energy)



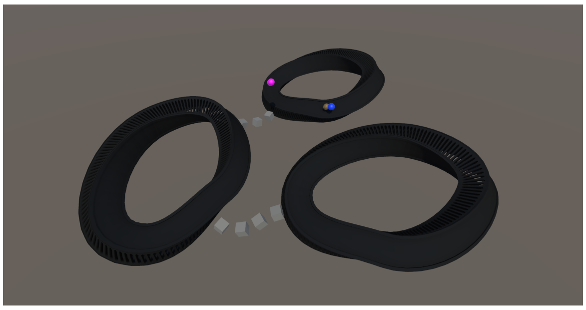
The character collects the energy item (+energy)



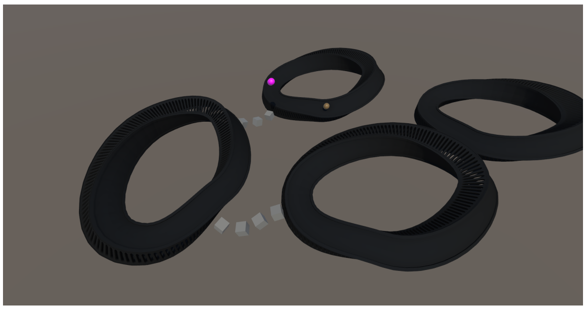
the player speed towards the next item



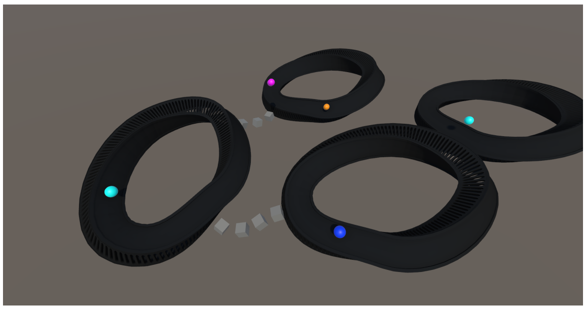
builds a new bridge (characters energy reduced)



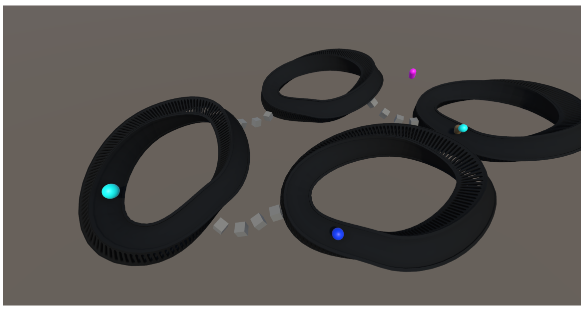
character collects new-shape item



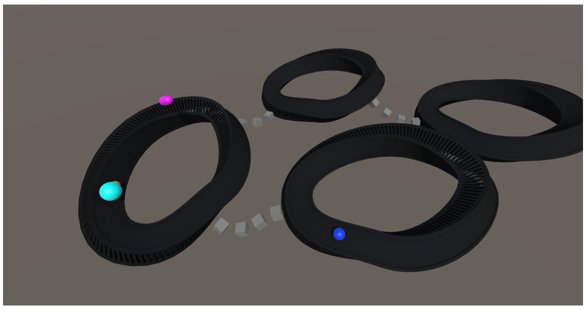
player places new shape



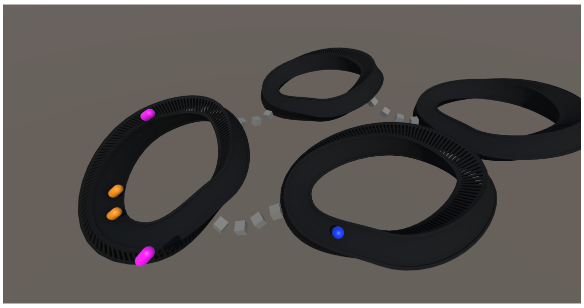
each shape checks whether it should spawn a new item, then 2 energy-items & 1 new-shape-item are spawned



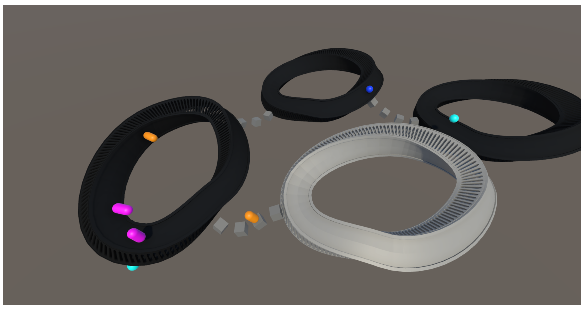
the player bridges and walks over to the energy-item



As the other energy item was around for some time, it became quite big (=much energy) before the player collects it



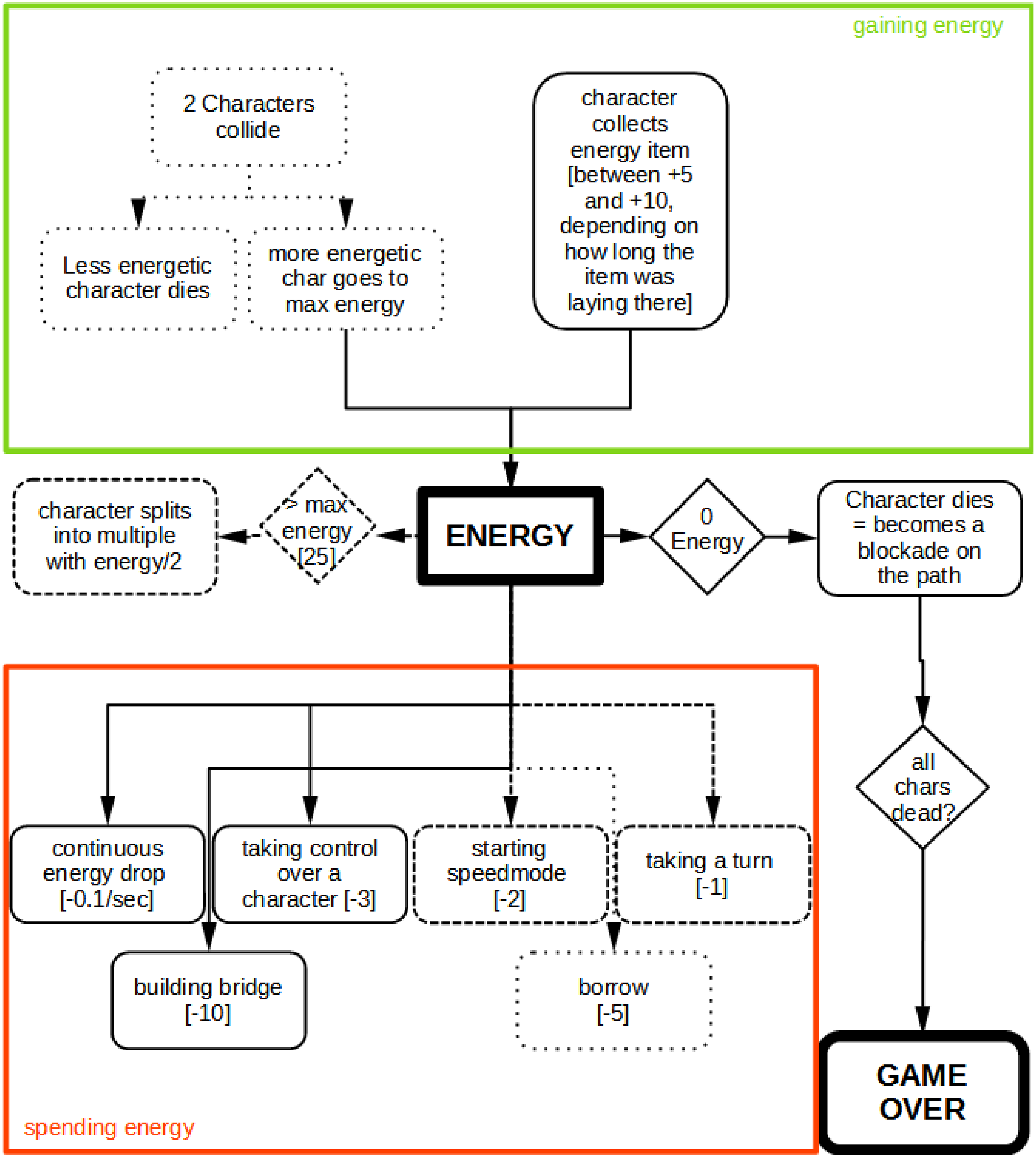
the over-energised character split into two character (see Energy-document)



the player was too slow at collecting the new-shape item -> the shape died and some new items spawned

Everything the player does in the game is based on energy

Each one of the characters has their energy, which is refilled / depleted by the following (inter)actions:



Gameplay Example: